# CLOSED-SUBGROUP CORRESPONDENCE IN INFINITE GALOIS THEORY

THE STACKS PROJECT AUTHORS

*Editorial note.* Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

*Allowed standard prerequisites (editorial).*

- Finite Galois theory, normal closures, profinite Galois topology and restriction-surjectivity for Galois extensions.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

*Source reference.* Original lemma 0BMJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14632–14632. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

*Source reference.* Original theorem 09DW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11879–11879. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

*Source reference.* Original lemma 0BMG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14629–14629. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

*Source reference.* Original lemma 0BMK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14633–14633. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

*Source reference.* Original lemma 09I2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12025–12025. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

**Theorem 0.1** (Fundamental theorem of infinite Galois theory)**.** *Let $L/K$ be a Galois extension. Let $G = Gal(L/K)$ be the Galois group viewed as a profinite topological group (Lemma 0BMJ). Then we have $K = L^G$ and the map*

$$\{\text{closed subgroups of } G\} \longrightarrow \{\text{subextensions } L/M/K\}, \quad H \longmapsto L^H$$

*is a bijection whose inverse maps $M$ to $Gal(L/M)$. The finite subextensions $M$ correspond exactly to the open subgroups $H \subset G$. The normal closed subgroups $H$ of $G$ correspond exactly to subextensions $M$ Galois over $K$.*

*Proof.* We will use the result of finite Galois theory (Theorem 09DW) without further mention. Let $S \subset L$ be a finite subset. There exists a tower $L/E/K$ such that $K(S) \subset E$ and such that $E/K$ is finite Galois, see Lemma 0BMG. In other

*Date*: stacks-project:fields.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

words, we see that $L/K$ is the union of its finite Galois subextensions. For such an $E$, by Lemma 0BMK the map $\mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$ is surjective and continuous, i.e., the kernel is open because the topology on $\mathrm{Gal}(E/K)$ is discrete. In particular we see that no element of $L \setminus K$ is fixed by $\mathrm{Gal}(L/K)$ as $E^{\mathrm{Gal}(E/K)} = K$. This proves that $L^G = K$.

By Lemma 09I2 given a subextension $L/M/K$ the extension $L/M$ is Galois. It is immediate from the definition of the topology on $G$ that the subgroup $\mathrm{Gal}(L/M)$ is closed. By the above applied to $L/M$ we see that $L^{\mathrm{Gal}(L/M)} = M$.

Conversely, let $H \subset G$ be a closed subgroup. We claim that $H = \mathrm{Gal}(L/L^H)$. The inclusion $H \subset \mathrm{Gal}(L/L^H)$ is clear. Suppose that $g \in \mathrm{Gal}(L/L^H)$. Let $S \subset L$ be a finite subset. We will show that the open neighbourhood $U_S(g) = \{g' \in G \mid g'(s) = g(s)\}$ of $g$ meets $H$. This implies that $g \in H$ because $H$ is closed. Let $L/E/K$ be a finite Galois subextension containing $K(S)$ as in the first paragraph of the proof and consider the homomorphism $c : \mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$. Then $L^H \cap E = E^{c(H)}$. Since $g$ fixes $L^H$ it fixes $E^{c(H)}$ and hence $c(g) \in c(H)$ by finite Galois theory. Pick $h \in H$ with $c(h) = c(g)$. Then $h \in U_S(g)$ as desired.

At this point we have established the correspondence between closed subgroups and subextensions.

Assume $H \subset G$ is open. Arguing as above we find that $H$ contains $\mathrm{Gal}(L/E)$ for some large enough finite Galois subextension $E$ and we find that $L^H$ is contained in $E$ whence finite over $K$. Conversely, if $M$ is a finite subextension, then $M$ is generated by a finite subset $S$ and the corresponding subgroup is the open subset $U_S(e)$ where $e \in G$ is the neutral element.

Assume that $K \subset M \subset L$ with $M/K$ Galois. By Lemma 0BMK there is a surjective continuous homomorphism of Galois groups $\mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$ whose kernel is $\mathrm{Gal}(L/M)$. Thus $\mathrm{Gal}(L/M)$ is a normal closed subgroup.

Finally, assume $N \subset G$ is normal and closed. For any $L/E/K$ as in the first paragraph of the proof, the image $c(N) \subset \mathrm{Gal}(E/K)$ is a normal subgroup. Hence $L^N = \bigcup E^{c(N)}$ is a union of Galois extensions of $K$ (by finite Galois theory) whence Galois over $K$. $\square$